Task 1A: Are these phenomena random? Give reasons for your answers. Not graded

(a) According to the definition: *Outcomes that we cannot predict but that nonetheless have a regular distribution in very many repetitions*.
- Temperature on a particular day at a specific time in a fixed location will NEVER be repeated
- We DO make probabilistic forecasts about such temperatures routinely
- …So what is the nature of the "repetition" when we do that? Weather systems – the planet! – do not "know" anything, and that includes a human scheme for marking time and date. The repetition is in the precursor conditions according to the weather forecasting model being used, which embodies our best understanding of the physical processes. It is those physical processes that we believe to be repetitive in a usable way.

(b) Assume the whole student body as sample space, and "you" selected at random. Then your id first digit would be random with a regular distribution – the distribution of first digits of those ids. Some schools begin all ids with a fixed number (755 or 33 or …) so the first digit would always be the same – a degenerate distribution. Some schools use other mechanisms.

Zachary Barnhart informs that the University of North Texas and the American Military University assign ids randomly.

(c) Yes: if we consider repeating the card selection (each time from a full deck), then the long run average frequency of any specific card will be 1/52. But the outcome of any single pick is unpredictable.

As you can see, what we consider a "random phenomenon" depends on how we choose to think about the context. And it is quite possible that a "random phenomenon" from one perspective is deterministic from another: random number generators are precisely of that nature.

Task 1B: Interpret the probabilities Not graded

(a) See the discussion for Task 1A.
(b) Here we naturally think of the probability of each valid digit – the frequency with which that first digit appears in the population of student ids. In the example of ids beginning with 755, the digit 7 would always occur. So, not random at all.
(c) 4/52 – the frequency of occurrence if "well shuffled" means that each card has the same chance of being selected.

Task 1C Independence Not graded

(a) Are specific dates, annually, independent as far as temperature is concerned? Not usually. Time of year is a consistent influence – summer temperatures being higher than winter temperatures. Multiyear cycles also occur. But, depending on the nature of the study, it may be reasonable to consider a set of such temperatures as independent around a given mean value.

(b) Are your Monday tweet counts independent? If you subscribe to a "service" that issues Monday tweets then a systematic factor exists. (Even if that is true, if the number is a negligible part of your total count, then it may be ignored.) For most people this is very likely an independent sequence. You can do this experiment (say, for the next 4 weeks) and examine the results…

(c) I hope not! The critical point here is the existence of a common factor ("YOU") and its importance in the effects of interest. Here is one factor: I may score well in Class A by spending a lot of my time studying for that class. As a result, I have less time to study Class B and Class C…so I am likely to score less well on those.

What do you think of that suggestion? (Another common factor: nature of the material in the classes…)

Several responses were of this form: *The grade I get in one course does not influence the grade I get in another course*. The way to think about probability is this: Does knowing the outcome for one course change what you think about the outcome for other courses?

Independence is often difficult to judge. It is frequently assumed as a starting point in analysis to simplify calculations. But it must be carefully assessed – a search for common underlying factors.

Task 2: 50 points

IPS 4.11 Probability Rules 30 points

(a) Addition rule (3) – disjoint events {HHH}, {TTT}. $\Pr = \frac{1}{8} + \frac{1}{8} = \frac{1}{4}$

(b) $\geq$ 1H and $\geq$ 1T in 3 tosses – this can occur in these 6 outcomes:
HHT, HTH, THH; HTT, THT, TTH;
Assuming equally likely outcomes the probability is 6 out of 8: formally using the addition rule as each of the individual sample outcomes is disjoint. Can we make that assumption here? The question does not give that information: it says only "A coin is tossed three times" and proceeds to give those two event probabilities**. So…no we cannot make that assumption. The correct approach is to express the event in terms of those two events for which we do know the probabilities: the complement of the event in (a). Hence probability 1-1/4=3/4.**
This is an easy question to get wrong because so much is familiar from the usual case of a fair coin and independent tosses. The lesson is the usual one: Read material carefully and explain your working.

(c) Pr(A)=0.5; Pr(B)=0.9; A,B disjoint.
This cannot happen because the probabilities of disjoint events add. The sum here would be greater than 1. Rule 3 for the sum of probabilities and Rule 1 (or 2) for the limit 1.

(d) Pr(A)= -0.04 Negative probabilities are not allowed. Rule 1.

(e) Complement – rule 4: $\Pr(A^C) = 1 - \Pr(A) = 1 - 0.23 = 0.77$

(f) 0, 1, or 2 heads are all the possibilities with two coin tosses.
So, probability = Pr(sample space)=1.This is Rule 2.

IPS 4.17 Distribution of blood types 10 points

(a) The required probability is $\Pr(O) = 1 - 0.42 - 0.11 - 0.03 = 0.44$ since the total probability is 1.

(b) We want $\Pr(0\ or\ A) = 0.44 + 0.42 = 0.86$. The outcomes are disjoint (a person cannot have more than one blood type) so the probabilities add.

IPS 4.18 Blood types in Ireland 10 points

Set up some convenient notation for events:
$\mathrm{US.O} \equiv$ US with blood type 0;
$\mathrm{I.O} \equiv$ Irish with blood type 0;

1. Now the probability we require is $\Pr\ (\mathrm{US.O}\ and\ \mathrm{I.O})$. Since the individuals are selected independently, the two events are independent and so the probabilities multiply.

$$\Pr(\mathrm{US.O}\ and\ \mathrm{I.O}) = \Pr(\mathrm{US.O}) \times \Pr(\mathrm{I.O}) = 0.44 \times 0.52 = \mathbf{0.2288}$$

2. $\Pr(same\ blood\ type) =$
$$\Pr\ ((US.A\ \&I.A) or\ (US.B\ \&I.B) or\ (US.AB\ \&I.AB) or\ (US.O\ \&I.O))$$

with an obvious extension of the notation…

$$= 0.42 \times 0.35 + 0.11 \times 0.10 + 0.03 \times 0.03 + 0.44 \times 0.52$$
$$= 0.147 + 0.011 + 0.0009 + 0.2288 = \mathbf{0.3877}$$

We use the independence of the US and Irish sample as in (1), and the fact that the four distinct combinations are disjoint (so the 'or' translates to simple '+'; there is no overlap and so no double counting to correct for).

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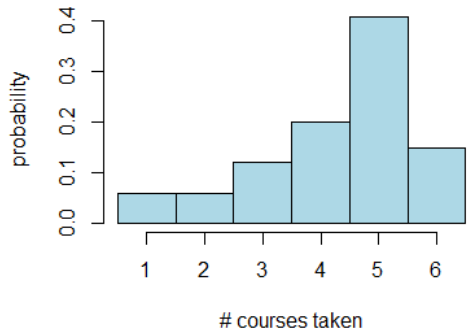
Task 3: IPS 4.37 Make a graphical display 5 points

Values that the random variable X can take on: 1,2,3,4,5,6
Probabilities with which those values occur: 0.06, 0.06, 0.12, 0.20, 0.41, 0.15.



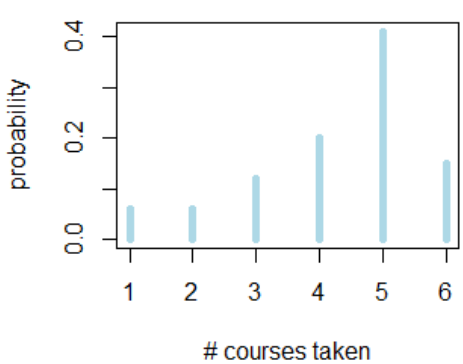
R

```
X     = 1:6
pcent = c(6,6,12,20,41,15)
prob  = pcent/100.0

## To use the hist() function, we need a sample variable
## (XX here) with the correct relative proportions for
## each outcome value. One way to do that is to create a
## vector of 5 ones, 5 twos, 13 threes, 26 fours, 36 fives
## and 15 sixes. That can be done using rep():

XX = rep(1:6,pcent)

## Here is a quick way to have the x-axis labels in the
## center of the displayed bars (see breaks=…)

hist(XX,breaks=0.5:6.5, probability=TRUE, col='lightblue',
  xlab="# courses taken", ylab="probability", main="")
```

There should be a title (or caption) telling the reader what the data in the graph represents. The data here refer to classes taken in the fall semester at an unnamed college – so we have to be somewhat vague. In practice, we would want to be as explicit as possible.

What do you think about the shape of this distribution? There is clear asymmetry – skew to the left. Unimodal. Do those details matter? Not in those technical words: Think about the meaning of the data. Number of courses taken. So we phrase the description in terms meaningful to that situation: The largest category is five courses, 41% of students. And so on.

NOTE When a random variable is discrete the histogram style above is strictly not correct. The bars at the explicit values that the random variable can assume should be distinct, separated by space as in a bar plot. That is to show that the intermediate values are not part of the domain of the random variable. For example, the value 1.2 is not a value that the random variable can assume here, yet the histogram does not show that. Here is a version that is considered more correct:



R

```
## Separating those bars

plot(table(XX)/100, type="h", lwd=5, col='lightblue',
  xlab="# courses taken", ylab="probability", main="")
```

Task 4 Nonstandard dice 10 points

Die 1: standard 6 sided die; Die 2: 1 on 3 faces and 6 on 3 faces.

Roll both dice once. The possible outcomes are 1,2,3,4,5,6 for die 1 and 1, 6 for die 2:

1				2
2				3
3	+	1	=	4
4				5
5				6
6				7

1				7
2				8
3	+	6	=	9
4				10
5				11
6				12

These outcomes are equally likely, so each has probability 1/12. Another way to see the result is this: the two dice are independent, so their outcomes are independent, so the probabilities of the outcomes multiply. For example:

$$\Pr(1\ and\ 1) = \Pr(1) \times \Pr(1) = \frac{1}{6} \times \frac{1}{2} = \frac{1}{12}$$

Here is a graphic display (borrowed from Ryan Rattazzi):

		Standard Die					
		1	2	3	4	5	6
Irregular Die	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	1	2	3	4	5	6	7
	6	7	8	9	10	11	12
	6	7	8	9	10	11	12
	6	7	8	9	10	11	12

Notice that the sum of the values has 11 possibilities: the value 7 can occur in 6 ways, while the other values, 1-6 and 8-12, can occur in 3 ways.

In summary:

Y	2	3	4	5	6	7	8	9	10	11	12
Pr(Y)	1/12	1/12	1/12	1/12	1/12	1/6	1/12	1/12	1/12	1/12	1/12